

Step 4 Results Note — Uncertainty Tagging and Confidence Calibration

Confidence scoring layered on reconstruction output, calibration analysis, and a validation-locked review threshold

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1. Purpose

Per the Stage 2 development plan, Step 4 layers confidence scoring on top of the reconstruction model's output and flags low-confidence spans as uncertain, so a human reviewer can focus attention on the reconstructions most likely to need correction. The flag is deliberately not a claim that a reconstruction is wrong: it means the model lacks enough evidence to return that word as reliable. Some correct words will be flagged, and some wrong words will slip through; the layer is judged on how well it concentrates real errors into the flagged set.

This note also closes an open question left from the reconstruction work: whether fine-tuned RoBERTa's small accuracy edge over the pretrained baseline is a real effect or noise. That is tested directly, with a bootstrap confidence interval, before the confidence layer is built on top of the reconstruction model.

2. Revisiting the Reconstruction Model's Accuracy Gain

The reconstruction model was retrained with `load_best_model_at_end` enabled, so the checkpoint kept is the one with the lowest validation loss rather than whichever epoch happened to finish last. Validation loss still rose after epoch 1 (2.622 → 2.630 → 2.691), the same pattern seen previously, and epoch 1 was correctly restored as the best checkpoint.

Metric	Pretrained RoBERTa	Fine-tuned RoBERTa (best checkpoint)
MLM loss (test set)	2.2367	1.6615
Top-1 accuracy, overall	48.17%	48.65%
Top-1 accuracy, content words	26.90%	27.03%

Table 1. Reconstruction performance after checkpoint selection by validation loss, on the full 1,478-example held-out test set.

Controlling for overfitting did not close the accuracy gap: it remained about half a point overall. To find out whether that half-point is a real effect, a paired bootstrap was run on the two models' per-example correctness over the same 1,478 test examples (10,000 resamples).

Quantity	Value
Observed difference (fine-tuned – pretrained)	+0.47 percentage points
95% bootstrap confidence interval	[−3.04, +4.06] percentage points

Table 2. Paired bootstrap on whole-word top-1 accuracy, held-out test set.

The interval includes zero, so this is not a statistically detectable improvement at this sample size. That is a narrower claim than “there is no effect”: the data are compatible with a true effect anywhere from roughly −3.0 to +4.1 percentage points, which covers both a small real advantage for fine-tuning and a small real disadvantage. The fine-tuned model is carried forward as the reconstruction engine for the confidence layer below because it is at least as accurate and has a substantially lower MLM loss, but its accuracy edge over the pretrained baseline should not be reported as established.

3. Method: Confidence Scoring

For a reconstructed word that decodes to k RoBERTa subword tokens, the model fills the k mask positions one at a time (the same iterative decoding used for evaluation), and at each step the probability it assigned to its own chosen token is recorded. The word's confidence score is the geometric mean of those k probabilities: $\exp(\text{mean}(\log p_1, \dots, \log p_k))$. An arithmetic mean or a plain product would understate confidence for longer words purely because they have more chances for a low-probability step; the geometric mean removes that length penalty so a two-token word and a one-token word are scored on a comparable scale.

4. Calibration Analysis (Validation Set, 2,000 Examples)

Confidence bin	Count	Avg. confidence	Actual accuracy
[0.0, 0.1)	97	0.066	0.021
[0.1, 0.2)	247	0.147	0.105
[0.2, 0.3)	249	0.249	0.141
[0.3, 0.4)	221	0.348	0.267
[0.4, 0.5)	171	0.450	0.386
[0.5, 0.6)	166	0.546	0.428
[0.6, 0.7)	139	0.642	0.547
[0.7, 0.8)	138	0.751	0.652
[0.8, 0.9)	157	0.850	0.752
[0.9, 1.0)	415	0.969	0.949

Table 3. Confidence calibration on the validation set. Expected Calibration Error (ECE) = 0.0704.

Accuracy rises monotonically with confidence bin, so the score is doing meaningful ranking: higher confidence reliably means higher odds of being correct. But the model is overconfident in every bin without exception, including the top one — average confidence 96.9% against actual accuracy 94.9% in the [0.9, 1.0) bin, a gap of about 2 points. The gap is wider through the middle of the range, roughly 4 to 12 points. This overconfidence is consistent enough across bins that a fixed threshold chosen on validation data can be expected to carry over to unseen data reasonably well, which the test-set results below confirm.

5. Threshold Selection (Validation Only)

The threshold was chosen using only the validation set, before the test set was touched. The rule: among confidence values attached to validation examples the model got wrong, take the lowest threshold at which flagging every example with confidence at or below it catches at least 80% of all wrong validation reconstructions — an error recall target set in advance, not fitted to a result. This threshold was then locked and applied to the test set exactly once, with no adjustment.

Quantity	Validation result
Locked threshold	0.5442
Error recall (share of wrong reconstructions flagged)	80.06%
Precision (share of flagged items that are wrong)	79.61%
Flag rate (share of all outputs flagged)	53.45%

Table 4. Threshold selection outcome on the validation set.

6. Locked-Threshold Evaluation on the Held-Out Test Set

Quantity	Test result
Error recall	77.21%
Precision	78.98%
Flag rate	50.20%

Table 5. Locked threshold applied once to the 1,478-example held-out test set, two unseen books.

The threshold transfers from validation to test with only minor degradation (error recall 80.06% → 77.21%, precision 79.61% → 78.98%), which is what a well-behaved, non-overfit threshold should look like.

6.1 The concrete benefit: what the unflagged output actually gives an archivist

Flag rate and precision describe the flagged half of the output. The more directly useful number is what happens to the other half — the reconstructions the system does not flag. With overall test accuracy at 48.65% (an error rate of 51.35%, or about 759 wrong words out of 1,478), and 50.20% of outputs flagged (about 742 words, of which 78.98% — about 586 — are genuinely wrong), the unflagged set contains about 736 words, of which about 173 are wrong.

Subset	Count	Error rate
All reconstructions (no flagging)	1,478	51.35%
Flagged ("review me")	~742	78.98%
Unflagged ("trust this")	~736	~23.5%

Table 6. Error rate before and after the confidence layer splits the output. Flagged/unflagged counts are derived from the recall, precision, and flag-rate percentages above.

That is the practical payoff for an archivist: trusting only the unflagged half of the output cuts the error rate from 51.35% to roughly 23.5%, less than half of the unfiltered baseline. The layer is not just pointing at likely problems; it is also certifying a majority-reliable subset that can be used with substantially more confidence than the raw reconstruction output.

7. Interpretation

The confidence layer works as intended on the dimensions that matter for this project. It ranks correctly (higher confidence means higher accuracy at every bin), it is moderately but consistently overconfident (ECE 0.0704, present in every bin), and a threshold chosen honestly on validation data alone transfers to unseen test data with only minor loss of recall and precision. Its practical effect is real: an unflagged reconstruction is roughly twice as likely to be correct as a random one from the unfiltered output.

The reconstruction model underneath it remains an open question rather than a settled one. Fine-tuning clearly improved MLM loss, but the accuracy advantage over pretrained RoBERTa is not statistically distinguishable from zero at this sample size (Table 2). The fine-tuned model was used here because it is not worse and trains to a better-behaved loss, not because its accuracy has been shown to be better.

8. Next Steps

- Proceed to Step 5: final end-to-end JSON integration, combining OCR region classification, whole-word reconstruction, and this confidence/uncertainty tagging into one unified per-document output, per the multi-task output design from the Stage 1 research brief.
- If future work revisits calibration, the consistent ~2–12 point overconfidence gap (Table 3) is a candidate for a recalibration step such as temperature scaling, applied after training rather than by changing the reconstruction model itself.
- If a fuller ablation is wanted later, the same calibration and threshold analysis could be run on the pretrained model to see whether its confidence signal is equally usable, given its accuracy is statistically indistinguishable from the fine-tuned model's.